



Bank Portal Authentication System

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Project Overview

Secure by Design

A robust authentication system built specifically for bank portal environments, protecting user data at every layer.

Multi-Layer Security

Combines JWT token management and Two-Factor Authentication for defence-in-depth security.

User-Friendly Flows

Intuitive registration and login workflows designed for clarity without sacrificing security.



Key Features



User Registration

Sign up with email and phone number for a verified account identity.

Email OTP Verification

One-time passcodes sent via email activate and validate each account.

Secure Login & Password Hashing

bcrypt-encrypted credentials ensure passwords are never stored in plain text.

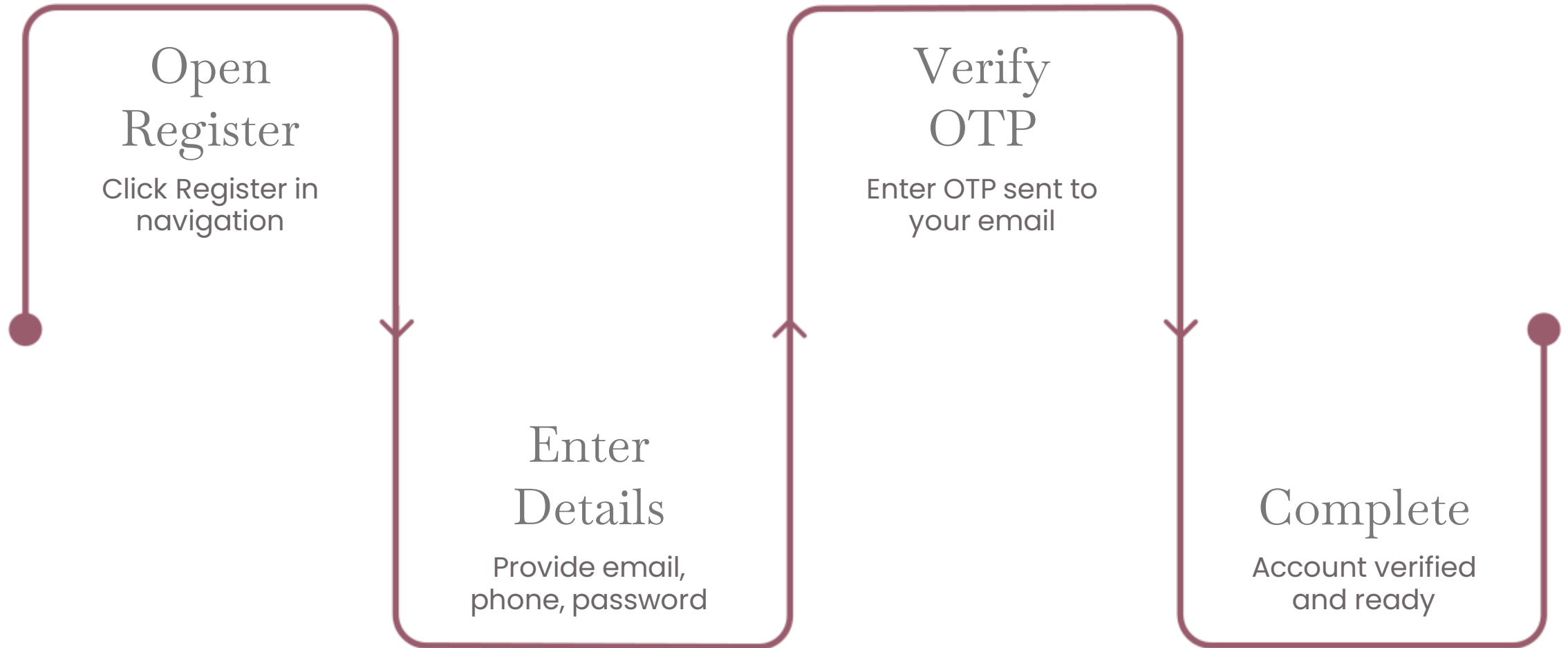
JWT Session Management

Stateless token-based sessions keep the portal fast and secure.

TOTP Two-Factor Authentication

Time-based one-time passwords add a critical second layer of protection.

Registration Process



A streamlined four-step process ensures every account is verified and secure before access is granted.

Login Process

Step-by-Step Login Flow

1 Navigate to Login

Click "**Login**" in the portal navigation bar.

2 Enter Credentials

Provide your registered **email address** and **password**.

3 2FA Verification

If 2FA is enabled, enter the **6-digit authenticator code**.

4 Access Dashboard

Gain secure access to your **personal bank dashboard**.



Two-Factor Authentication Setup

1

Prompted After Login

System prompts 2FA setup on first successful login.

2

Scan QR Code

Use any authenticator app (Google, Authy) to scan the displayed QR code.

3

Verify 6-Digit Code

Enter the time-based code generated by your authenticator app.

4

2FA Active

All future logins now require the authenticator code for access.

❏ TOTP codes refresh every 30 seconds — ensuring no code can be reused by an attacker.



Authentication Features

Email OTP Verification

Secure account activation through one-time passcodes delivered to the registered email.



Password Hashing

bcrypt encryption ensures all passwords are salted and hashed — never stored in plain text.



JWT Tokens

Stateless session management via signed JSON Web Tokens for scalable, secure access control.



TOTP 2FA

Time-based one-time passwords provide a dynamic second factor that expires every 30 seconds.

Backend Technologies



Powering the Secure Backend



Node.js Runtime

High-performance JavaScript runtime for handling concurrent authentication requests.



Express.js Framework

Lightweight and flexible REST API framework powering all authentication routes.



JWT · bcrypt · Speakeasy

Industry-standard libraries for token signing, password hashing, and TOTP generation.



Nodemailer

Reliable email delivery service for dispatching OTP verification codes to users.

Frontend Technologies



HTML₅

Semantic markup delivering accessible, well-structured portal interfaces.



CSS₃

Modern styling for a clean, responsive bank portal experience across devices.



JavaScript ES6+

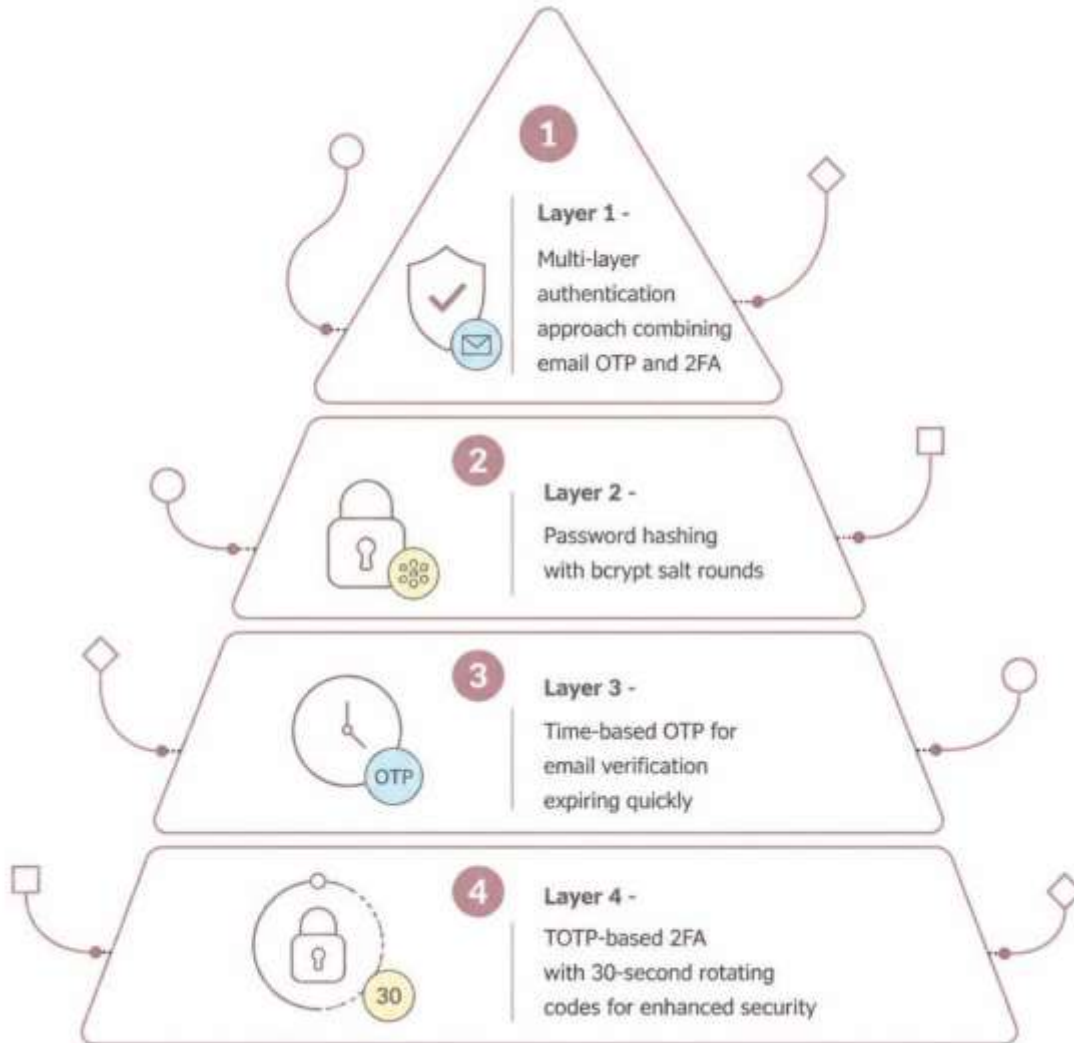
Dynamic client-side scripting enabling real-time validation and interactive flows.



QR Code Library

The *qrcode* library renders TOTP QR codes for seamless authenticator app setup.

Security Highlights & Benefits



Why This Architecture?

The Bank Portal Authentication System is built on **four complementary security layers**, each addressing a distinct threat vector:

- **Multi-layer authentication** ensures no single point of failure exists in the login flow.
- **bcrypt password hashing** makes brute-force attacks computationally infeasible.
- **Time-limited OTPs** for email verification prevent replay attacks.
- **TOTP 2FA** guarantees that even stolen credentials cannot grant unauthorized access.

Together, these layers deliver **enhanced security**, a **seamless user experience**, and **industry-standard practices** — with a scalable architecture ready for future enhancements.

Thank You

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BANK PORTAL AUTHENTICATION SYSTEM

ACADEMIC PROJECT PRESENTATION

